Bus Reservation And Ticketing System

https://github.com/aiyshwary/Bus-Reservation-and-Ticketing-System

Java | OOP | Console Application

OVERVIEW

A Java-based console application that simulates a complete bus ticket reservation and billing workflow, supporting multi-passenger booking, seat management, and discount-based billing operations.

IMPACT

Built a fully functional console-based Bus Reservation and Ticketing System in Java with structured input validation, in-memory seat management, and automated billing with passenger-type discounts.

TECHNICAL WALKTHROUGH

"I built a console-based Bus Reservation and Ticketing System in Java that
allows seat management, passenger booking, billing with discounts, and transaction tracking using structured input validation and in-memory data
storage. "

Problem statement
Manual ticket booking is error-prone and slow.
This project simulates a real-world bus reservation system where users can manage destinations, seat availability, passenger bookings, billing, and payment status through a menu-driven interface.

High-level system flow

Explain this in steps

Step 1: Authentication

Simple login using username and password

Prevents unauthorized access
"This simulates role-based access at a basic level."

Step 2: Main Menu

The system has 5 main modules

- **i)** Destination Management
- **ii)** Passenger Booking
- **iii)** Billing & Payment
- **iv)** View Passenger Details
- **v)** Exit

Module-wise explanation

1. Destination Module

- **i)** Displays available destinations

Shows

- **i)** Fare per destination
- **ii)** Remaining seat count
- **iii)** Seats are initialized dynamically

“Seat availability updates in real time after each booking.”

2. Passenger Booking Module

This is the core logic.

What happens here

- **i)** Passenger name validation (no duplicates)
- **ii)** Destination selection with validation
- **iii)** Seat availability check
- **iv)** Passenger count input
- **v)** Discount handling
- **vi)** Fare calculation

Business logic

- **i)** Total Fare =
- **ii)** (Regular passengers × fare)
- **iii)** + (Discounted passengers × (fare – 20%))

“This ensures accurate billing while enforcing seat limits.”

3. Billing & Payment Module

i) Search passenger by name

ii) Prevent double payment

iii) Accept payment

iv) Compute change

v) Mark passenger as PAID

“Once paid, the passenger status is locked to avoid duplicate billing.”

4. View Module

i) Search passenger details

Displays

i) Destination

ii) Passenger count

iii) Discount count

iv) Total fare

v) Payment status

vi) Amount paid and change (if applicable)

“This acts like a transaction history lookup.”

5. Exit Module

Gracefully exits the system

Confirms continuation

KEY DETAILS

1. Implements user authentication, seat selection, and multi-passenger booking in a single session.

2. Handles billing logic including discount tiers for different passenger categories.

3. Uses structured in-memory data storage for transaction tracking without a database dependency.

4. Validates all inputs to prevent double-booking and illegal seat selections.